

Alaa Anani

Saarbrücken, Germany

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[AlaaAnani](#) • [Google Scholar](#)

Education

Max Planck Institute for Informatics

CISPA Helmholtz Center for Information Security

Ph.D. in Computer Science

Area: Trustworthy machine learning.

Supervisors: [Prof. Dr. Bernt Schiele](#) and [Prof. Dr. Mario Fritz](#)

Saarbrücken, Germany

Aug 2024 – Present

Saarland University

M.Sc. in Computer Science – GPA: 1.6 (Best is 1.0)

Thesis: Adaptive Hierarchical Certification for Segmentation using Randomized Smoothing

Supervisors: [Prof. Dr. Bernt Schiele](#) and [Prof. Dr. Mario Fritz](#)

Saarbrücken, Germany

Oct 2021 – May 2024

The American University in Cairo

B.Sc. in Computer Engineering – GPA: 3.96/4.0 (Summa Cum Laude), Minor: Mathematics

Thesis I: Mixture of Experts for Human Activity Classification using Semantic Fusion

Thesis II: On Ramsey Theory and its Computer Science Algorithmic Applications

Supervisors: [Prof. Cherif Salama](#), [Prof. Mohamed Moustafa](#) and [Prof. Daoud Siniora](#)

Cairo, Egypt

Sept 2016 – June 2021

Publications

[1]: **A. Anani**, T. Lorenz, B. Schiele, M. Fritz, and J. Fischer. “Certified Circuits: Stability Guarantees for Mechanistic Circuits.” In *International Conference on Machine Learning (ICML)*, 2026.

[2]: **A. Anani**, T. Lorenz, M. Fritz, and B. Schiele. “Pixel-level Certified Explanations via Randomized Smoothing.” In *International Conference on Machine Learning (ICML)*, 2025.

[3]: **A. Anani**, T. Lorenz, B. Schiele, and M. Fritz. “Adaptive Hierarchical Certification for Segmentation using Randomized Smoothing.” In *International Conference on Machine Learning (ICML)*, 2024.

Research Experience

SpreeAI

Machine Learning Research Engineer

Topics: 3D head tracking, portrait illumination transfer, face texture transfer, and head restoration

Reference: [Dr. Minh Vo](#)

Los Angeles, USA

Oct 2021 – Dec 2023

École Polytechnique Fédérale de Lausanne – MLO Lab

Summer@EPFL Intern

Topic: Interpretability of a COVID-diagnosis ViT from lung ultrasound images

Reference: [Dr. Mary-Anne Hartley](#)

Lausanne, Switzerland

July 2021 – Oct 2021

Dell Technologies – Data Office

Undergraduate Data Science Research Intern

Topics: Technology forecasting and organized egypt’s first AI hackathon ([C4F2](#))

Reference: [Eng. Steve Todd](#)

Cairo, Egypt

Dec 2020 – June 2021

Synapse Analytics – Data Science Team

Data Science Intern

Topic: Credit scoring with time-series analysis

Cairo, Egypt

July 2020 – Aug 2020

The American University in Cairo

Undergraduate Researcher

Topic: Deep reinforcement learning for traffic intersection control

Reference: [Prof. Cherif Salama](#)

Cairo, Egypt

Sept 2019 – June 2020

Project Highlights

Practical Machine Deep Learning

Implemented CNN from scratch using python

([GitHub](#))

Fall 2020

AUC Robotics Club

Full software system for a remotely operated underwater vehicle [Vader I and II]

Spring 2020

Distributed Systems

A P2P photo-sharing system using UDP

([GitHub](#))

Fall 2019

Computer Architecture

3-Stage pipe-lined CPU with interrupts support

Fall 2018

Teaching

Saarland University & MPI for Informatics

Graduate Teaching Assistant

Course: Automated Reasoning

Reference: [Prof. Uwe Waldmann](#)

Saarbrücken, Germany

Oct 2021 – March 2022

The American University in Cairo

Undergraduate Teaching Assistant

Course: Graph Theory

Reference: [Prof. Daoud Siniora](#)

Cairo, Egypt

Feb 2021 – June 2021

The American University in Cairo

Undergraduate Teaching Assistant

Course: Discrete Mathematics

Reference: [Prof. Daoud Siniora](#)

Cairo, Egypt

Feb 2021 – June 2021

The American University in Cairo

Undergraduate Teaching Assistant

Course: Embedded Systems Lab

Reference: [Dr. Suzanne Safwat](#)

Cairo, Egypt

Feb 2021 – June 2021

Awards

2025: Best Poster Award, ELSA General Assembly, Belgium

[European Lighthouse of Secure and Safe AI](#)

2024: Best Poster Award, ELSA General Assembly, UK

[European Lighthouse of Secure and Safe AI](#)

2021: CS@Max Planck Highly Selective Doctoral Grant

[CS@Max Planck](#)

2021: Dr. Abdel Rahman El Sawy Award in Engineering (\$1K)

[AUC](#)

2021: 4th Place, WomenHack Data Science Hackathon

[McKinsey](#)

2019: Top-10 ROV Team in the Middle East

[MATE ROV Regional Competition](#)

2019: 4th Place in the Minesweepers Competition

[Minesweepers](#)

2016–2021: Dean's List for Academic Standing

[AUC](#)

2016: Egyptian Government PSSF Scholarship (~\$80K)

[Egyptian Government](#)

Extracurricular

2017–2020: Robotics control team leader in the ROV competition

[MATE ROV](#)

2018: Academic Director

[AUC Robotics](#)

Skills

Programming: Python, C, C++, CUDA C, ROS, Verilog, MATLAB, LabVIEW

Frameworks: PyTorch, CUDA, TensorFlow, Docker, Kubeflow, UNIX, Gazebo, SUMO, $\LaTeX$